Causal-Intervened Contrastive Learning With Collaborative Domain Alignment for Cross-Domain Few-Shot Hyperspectral Image Classification

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***Abstract*—Cross-domain few-shot hyperspectral image (HSI) classification aims to classify land cover categories in a target domain (TD) with scarce labels by transferring knowledge from a source domain (SD). However, the inherent spectral variability among different scenes often introduces spurious correlations, decreasing the generalization ability of few-shot models. In this article, we propose a novel causal-intervened contrastive learning with collaborative domain alignment (CCCDA) framework that mitigates these spurious correlations and enhances cross-domain feature alignment. Specifically, we propose a causal intervention-guided contrastive learning (CICL) module, which presents a random mask causal-intervened (RMCI) strategy to generate counterfactual samples by preserving class-related central regions while disturbing noncausal backgrounds. Subsequently, an intervention-invariant contrastive learning process is designed to minimize the feature distance between query samples and their counterfactual counterparts, enforcing the model to learn causally invariant features. Furthermore, to overcome the domain gap, we propose the collaborative domain alignment (CDA) module to integrate a causal meta-learning branch, which applies prototype matching on support and mask sets, with a conditional adversarial training branch that aligns domain-invariant features between the SD and TD. Extensive experiments on four publicly available target HSI datasets demonstrate that the proposed CCCDA framework is superior to other state-of-the-art (SOTA) methods. The code is available at https://github.com/Chirsycy/CCCDA**

***Index Terms*—Causal intervention, cross-scene, domain adaptation, feature alignment, few-shot learning (FSL), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL image (HSI) captures data in hundreds of contiguous spectral bands at each spatial location, forming a 3-D data cube structure, with abundant spatial and spectral information [1]. This structure enables HSI to gain important information through rich spectral representation for land class recognition. Among HSI interpretation, HSI classification is a critical task that aims to assign a unique class label to each pixel of the HSI by identifying spatial–spectral feature of the object [2]. Nowadays, HSI classification is widely applied for land cover classification [3], vegetation cover analysis [4], and the prediction of soil properties [5].

In recent years, deep learning frameworks have flourished in the direction of HSI classification [6] through powerful feature extraction capabilities, such as convolutional neural networks (CNNs) [7], [8], and graph neural networks (GNNs) [9], [10]. Makantasis et al. [11] proposed a deep learning classification method that combined CNN and multi-layer perceptron (MLP) to construct high-level features of spatial–spectral information of the HSI pixels for accurate HSI classification. Li et al. [12] proposed a 3D-CNN HSI classification framework for simultaneously extracting spatial–spectral features, which captured local 3-D patterns that helped boost the classification performance. Roy et al. [13] employed 3-D and 2-D CNN networks by different morphology convolution kernels, which combined complementary spatial and spectral information to extract semantic features for HSI classification. Besides, Hong et al. [9] developed a minibatch graph convolutional network (GCN) that combined CNNs and GCNs to extract different types of HSI features, and the minibatch GCN yielded low computational cost and stable local optima. Due to the limitations of CNNs in capturing long-range dependencies, CNNs fall short in fully exploiting the rich spatial–spectral information in HSI. The transformer architecture [14], [15] facilitated the effective capture of long-range spatial–spectral information in HSI classification through an efficient global feature extraction capability. A notable example was the spectral–spatial transformer network (SSTN) proposed by Zhong et al. [16], which consisted of spatial attention and spectral association modules to overcome the limitations of convolutional kernels and improve overall classification performance. Furthermore, the integration of graph attention with transformer architectures [17] has shown great potential in progressively refining features for complex HSI tasks. Although the above approaches improved the accuracy of HSI classification, the performance of the approaches often degraded when applied to new HSI datasets acquired under different conditions, which is known as domain shift [18], [19].

Cross-domain HSI classification aims to transfer knowledge from a well-labeled source domain (SD) to a target domain (TD) with limited annotations, removing the challenge of domain shift [20], [21], [22], [23]. In cross-domain HSI classification, researchers explored deep learning frameworks

Received 21 May 2026; revised 21 August 2026; accepted 31 August 2026. Date of publication 7 September 2026; date of current version 17 September 2026. The work of Chunyan Yu was supported by the National Natural Science Foundation of China under Grant 62471079. *(Corresponding author: Chunyan Yu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3731169

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combined with various domain-adaptive (DA) methods to learn domain-invariant representations of spatial–spectral features [24]. Lee et al. [25] proposed a cross-domain CNN where the shared portion of the network was trained using multiple HSI datasets to address cross-domain HSI classification. Guan and Lam [26] proposed a cross-domain contrastive learning framework, which extracted domain-invariant information with a cross-domain task to reduce the issue of label scarcity. Similarly, Kong et al. [27] developed a prototype-guided momentum contrastive learning method to further enhance feature representation in cross-domain scenarios. Zhang et al. [28] developed the single-SD expansion network (SDEnet), which employed a generative adversarial network to synthesize an extended domain from a single source to improve generalization to unseen TDs. Li et al. [29] proposed a two-stage domain adaptation method that aligned domains adopting the maximum mean discrepancy criterion and proposed a spatial–spectral Siamese network to learn a more discriminative feature space. The above methods made progress in relieving the domain shift problem, but the SD and TD often originated from different geographical areas [30], [31], and the scarcity of labeled samples in the TD also exacerbated the problem of domain shift [32], [33].

Cross-domain few-shot HSI classification aims to transfer knowledge from a well-labeled SD to effectively classify data in a sparsely labeled TD with a different data distribution [34], [35], [36]. Researchers explored combining few-shot learning (FSL) [37], [38], [39] with DA [40], [41], [42], [43] to address the dual challenges of domain shift and label scarcity. Wang et al. [44] proposed a dual-branch domain adaptation FSL network that simultaneously performed domain fusion for global alignment and domain separation to address class distribution distortion caused by domain alignment in cross-domain HSI classification. Zhang et al. [45] proposed the Gia-cross-domain few-shot learning (CFSL) framework that adopted graph networks to model intradomain and interdomain relationships, handling issues of nonlocal spatial information and new classes in the TD. To address the issue of varying conditional distribution discrepancies in classes, Ye et al. [46] presented the adaptive domain-adversarial FSL (ADAFSL) framework for cross-domain HSI classification, which assigned different weights for each class based on a calculated discrepancy score, focusing on classes with large domain shift. In addition, Zhang et al. [47] proposed a spectral–spatial distribution consistent (SSDC) network based on meta-learning, which improved HSI domain adaptation by performing parallel alignment on spectral features and on spatial features. Li et al. [48] proposed a few-shot classification framework with SSL that enabled few-shot HSI classification by transferring knowledge from natural images, utilizing a self-supervised task to learn invariant features. Wang et al. [49] proposed the graph meta-transfer network (GMTN), which integrated GraphSAGE with meta-learning to mitigate information interference in few-shot HSI classification where domains have different classes. Despite the progress of existing methods, effectively transferring models for few-shot HSI classification remained a challenge.

Due to the variations in illumination, atmospheric conditions, and sensor properties across different scenes, the phenomenon of spectral variability is severe in cross-scene classification with few shots, and the obtained feature becomes statistically highly correlated with domain-level appearance factors, rather than reflecting the intrinsic class-related characteristics. The correlations between feature and labels are not intrinsic properties of the land-cover categories and fail to generalize among diverse domains. Such class-irrelevant correlations mislead the HSI classification model into capturing statistical information rather than stable and domain-invariant information. In our article, we refer to the mentioned correlation as a spurious correlation, which leads the model to rely on noncausal information and thus degrades performance on the TD.

To address the challenges of domain shift and spurious correlations in cross-domain few-shot HSI classification, we propose a novel causal-intervened contrastive learning with collaborative domain alignment (CCCDA) framework. The proposed CCCDA consists of two main stages: causal intervention-guided contrastive learning (CICL) and collaborative domain alignment (CDA). First, the CICL stage incorporates a random mask causal-intervened (RMCI) module to generate counterfactual samples and an intervention-invariant contrastive learning branch to ensure feature invariance against noncausal disturbances. Second, the CDA module is employed to achieves robust global domain alignment and knowledge transfer through causal meta-learning and conditional adversarial training.

In summary, the main contributions of this article can be summarized as follows.

1) We propose an innovative CCCDA framework for cross-domain few-shot HSI classification. By integrating a causal intervention mechanism into the meta-learning framework, our model generates a mask set to enhance causal feature representation. Through the synergy of intervention-invariant contrastive learning and CDA, the framework mitigates the negative effects of spurious correlations caused by spectral variability and guarantees effective knowledge transfer from SD to TD.
2) We propose a CICL strategy, which consists of the RMCI module and an intervention-invariant contrastive learning branch. The RMCI module generates counterfactual samples by preserving the causal center and randomly masking the noncausal background, providing a mask set that structurally enhances the meta-learning framework. Based on this, the intervention-invariant contrastive learning branch utilizes a mask-weighted loss to maximize the similarity between original query samples and their counterfactual views, forcing the feature extractor to focus on stable, causal-related features.
3) We propose the CDA module to achieve robust domain alignment through a dual-branch training mechanism. The causal meta-learning branch performs prototype-based classification on the support and mask sets to enhance class-level discrimination. Simultaneously, the conditional adversarial training branch utilizes a domain discriminator to align the conditional feature

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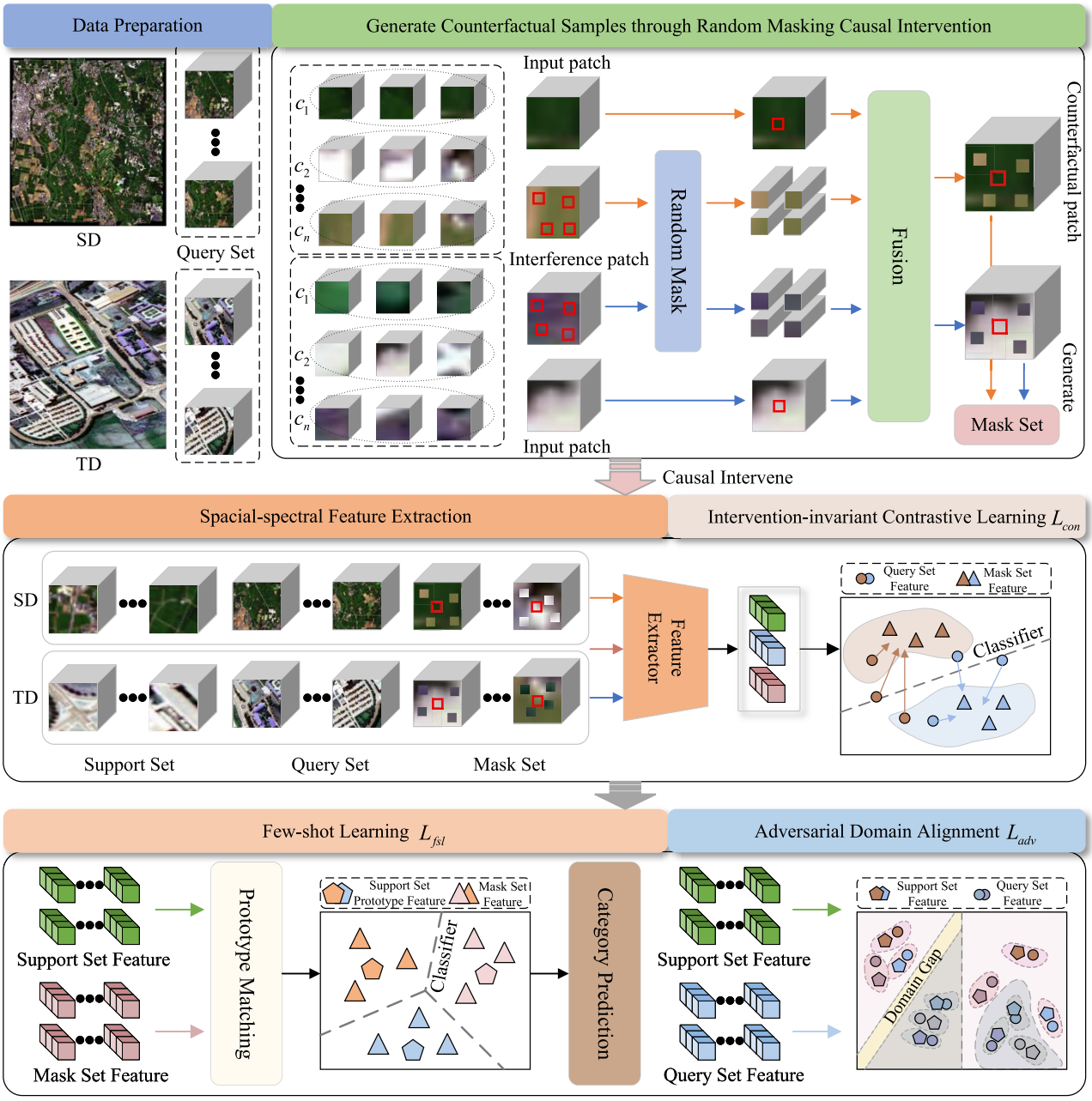


Fig. 1. Overview of the proposed CCCDA framework. Initially, the original HSI data are fed into the preprocess module to obtain preliminary features. Afterward, the RMCI module generates a counterfactual mask set from the original query set through causal intervention, which structurally enhances the meta-learning framework. Furthermore, the support set, query set, and the generated mask set are fed into the feature extractor to obtain corresponding feature representations. Finally, the framework is optimized through a multifaceted training strategy. The intervention-invariant contrastive learning branch utilizes the query and mask sets to minimize the contrastive loss (L_{con}), enforcing feature invariance. Meanwhile, the CDA module employs conditional adversarial training (L_{adv}) to bridge the domain gap. Notably, to address the few-shot classification task, a prototype matching strategy is adopted to compute class prototypes from the support set and predict labels for the mask set based on Euclidean distance, yielding the classification loss (L_{fsl}).

distributions of the SD and TD. Cooperatively, these branches reinforce the causal connection between features and labels while bridging the distribution gap across domains.

II. Proposed Method

Fig. 1 illustrates the overall architecture of the CCCDA. Initially, the original HSI data are divided into support sets and query sets, and the input samples are processed by the preprocess module to extract preliminary features. Afterward, the query sets are fed into the RMCI module of the CCCDA for generating counterfactual samples, which is accomplished by preserving the causal information with the center of the class-related region and disturbing with surrounding noncausal information. All counterfactual samples are constructed as a mask set that structurally enhances the meta-learning framework. Furthermore, the support set, query set, and obtained

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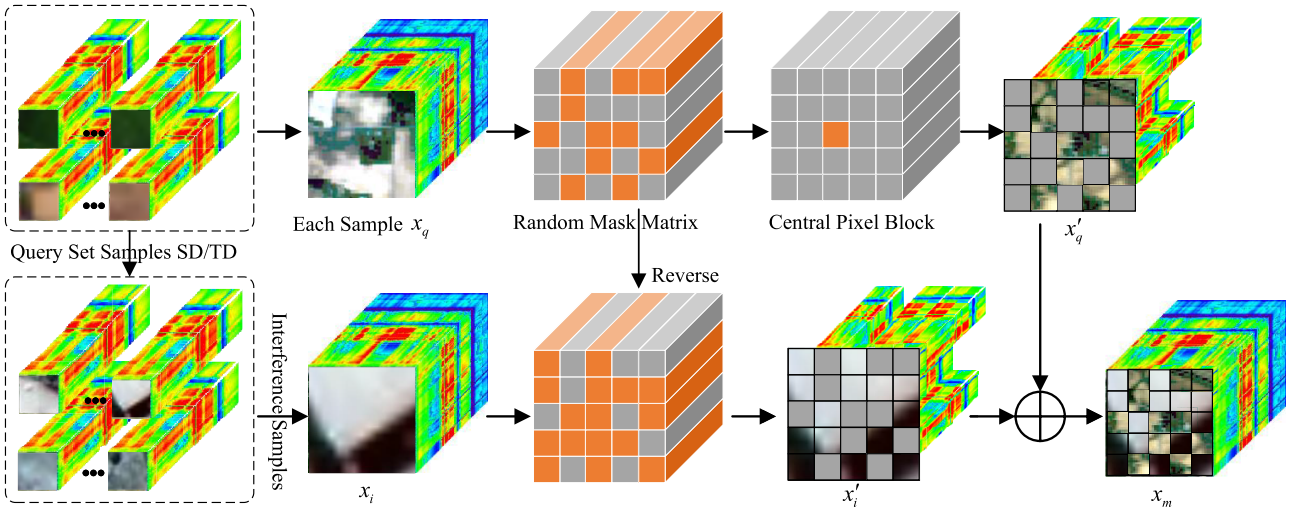


Fig. 2. Schematic of the RMCI module. The module generates a counterfactual sample x_m by mixing a query sample x_q with an interference sample x_i. A random mask matrix is applied to x_q to preserve the causal central pixel block while masking parts of the background. Next, the mask is reversed and applied to x_i to introduce noncausal features from a different class. The final output x_m is synthesized by summing the preserved causal features of x_q and the noncausal features of x_i.

mask set are fed into the backbone to obtain the corresponding feature representations. Finally, the obtained features are processed by the CDA module to achieve causally-invariant alignment and enhance causal connections between the features and labels. Notably, the classification loss of the FSL (L_{fsl}) is calculated by applying the prototype matching strategy in the features from both the support and mask sets. The loss of the contrastive training branch (L_{con}) and the loss of the adversarial training branch (L_{adv}) are gained by the CDA module.

A. Problem Definition

In the setting of cross-domain FSL, we have two datasets: the SD dataset that contains sufficient labeled samples for C_s classes and the TD dataset with a limited number of labeled samples for C_t classes. The TD dataset can be divided into two subsets: the training data D_{tra} with labeled data and the test data D_{tes} with unlabeled data, $D_{tra} \cup D_{tes} = D_t$, $D_{tra} \cap D_{tes} = \emptyset$. Following the common few-shot setups, each FSL task T consists of a support set (S) and a query set (Q), where S is a labeled set, and Q is an unlabeled set. When making a meta-task, we sample them randomly from D_s or D_t and S is selected in a C-way K-shot manner, where the S contains K labeled samples for each of C classes, which is denoted as an episode. The Q contains samples from the same classes with S in a meta-task. FSL is accomplished by alternating training on SD tasks with TD tasks. In the case of the SD episodic training procedure, the SD support set $S_s = \{(X_i, y_i), y_i \in C_s\}_{i=1}^{C\times K_1}$ is composed of randomly selected C classes with K_1 samples, and the SD query set has k_2 samples $Q_s = \{(X_j, y_j), y_j \in C_s\}_{i=1}^{C\times K_2}$. The TD episodic training procedure is the same as before. In our article, the main objective is to train the framework that leverages the extensive knowledge from the label-rich D_S and the few information from the labeled samples in D_{tra} to overcome the domain shift in the different HSI domain and the scarcity of labeled data in the TD. The performance of the framework is evaluated by accurately classifying the unlabeled samples in Q_{tes}.

B. Causal Intervention-Guided Contrastive Learning

In this article, we propose a CICL based on counterfactual inference principles [50], which consists of an RMCI module and an intervention-invariant contrastive learning process.

1) Implementation of RMCI: In HSI classification, training samples are typically constructed as small patches and the patch label is assigned according to the central pixel. The labeling strategy implicitly acknowledges that the central neighborhood has primary causal relevance for predicting the label, while the other surrounding pixels are susceptible to interference from neighboring categories. Following the stated criteria supplied by Hu et al. [51], we designed the following causal intervention scheme for the HSI cross domain classification with few shot. First, the causal intervention should keep semantic and not be overly destructive. Second, the causal intervention must be moderate and preserve the original data distribution. In the RMCI module, we keep the center pixel unchanged to avoid label corruption. Meanwhile, the surrounding neighborhood is highly correlated with the center pixel, thus, we adopt a fixed $\delta \times \delta$ region as the local causal region, mitigating overfitting from relying solely on the center pixel. Besides, we inject only a small portion of pixels sampled from other classes within the same dataset, rather than replacing large regions or introducing out-of-distribution content. In this way, we preserve distribution consistency by using pixels from the same domain as much as possible. Moreover, randomly selecting the number of replacements also keeps the intervention appropriately moderate. Initially, as illustrated in Fig. 2, for each query sample x_q with label l_q from SD or TD ($x_q \in Q_s/Q_t$), we randomly select an index i_{rand} from the candidate set $I_{cand} = \{j \mid L_q[j] \neq l_q\}$ to obtain an interference sample x_i. Afterward, x_q and x_i are fed into the RMCI module and the RMCI generates a random mask matrix $m^{H\times W} \in \{0, 1\}$, where the central $\delta \times \delta$ pixels are

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fixed to 1, and a fraction σ of the remaining positions are randomly set to 0 to mask the background information for subsequent interference. The input sample x_q is multiplied by m to obtain the premixed image x'_q. Furthermore, m is inverted and multiplied with x_i to obtain the mixing pixel x'_i at the corresponding position of x'_q. Finally, the counterfactual sample x_m is obtained by mixing x'_q and x'_i, which is defined as (1). All counterfactual samples constitute the mask set (M)

$$x_m = x_q \odot m + x_i \odot (1 - m) \tag{1}$$

where $\odot$ represents the Hadamard product. Each element $m_{i,j}$ of the mask matrix is calculated as follows:

$$m_{i,j} \begin{cases} 1, & if \quad (i,j) \in center \\ 1, & if \quad (i,j) \notin center \quad and \quad r > \sigma \\ 0, & if \quad (i,j) \notin center \quad and \quad r \le \sigma \end{cases} \tag{2}$$

where *center* denotes the fixed central $\delta \times \delta$ region, r is a random sampled value from the uniform distribution of [0, 1], and σ is the masking ratio of the pixels.

We have experimented that when the interference ratio σ is 25%, 50%, and 75%, respectively, random interference 50% can significantly improve the precision of the test set, which proves the effectiveness of the RMCI module. Please refer to Section III-C for details. The implementation of RMCI is presented in Algorithm 1.

2) Intervention-Invariant Contrastive Learning: After the feature extraction process, we obtain support set features S, query set features Q, and mask set features M, respectively. With the Q and M of the meta-tasks from SD or TD, the causal contrastive training branch is presented to minimize the feature distance for samples of the positive class while maximizing the distance for samples of negative classes. In addition, to effectively learn representations that are invariant to causal interventions, we propose a mask-weighted intervention-invariant contrastive loss. The loss function extends the standard supervised contrastive learning framework by not only treating samples from the same class as positives, but also defining each original query sample and its corresponding counterfactual version as a strong positive pair. Notably, the loss mechanism is designed to be aware of the interventional magnitude of the RMCI, which leverages the M to compute a per-sample invariance score that quantifies the proportion of unmodified pixels. Next, the score weights the similarity of the causal pair within the numerator of the contrastive loss, which enhances the reliance of the feature space on the stable, causal-related features preserved during the intervention. In this way, the causal contrastive training branch mitigates the impact of spurious correlations by shifting the learning objective of the model from observational probability $p(Y|X)$ to causal relationship $p(Y|do(X))$. The loss of contrastive training is defined as follows:

$$L_{con} = \frac{1}{N} \sum_{i=1}^{N} L_i \tag{3}$$

Algorithm 1 RMCI Algorithm

Input:
The batch of feature patches X_q of size (B, C, H, W);
The batch of corresponding labels L_q;
The masking ratio σ;
Output:
The batch of counterfactual patches X_m;
The batch of generated masks M_{out};

```
for k = 1 to B do
    x_q ← X_q[k];
    l_q ← L_q[k];
    I_cand ← indices {j | L_q[j] ≠ l_q};
    i_rand ← RandomSelect(I_cand);
    x_i ← X_q[i_rand];
    Generate mask m_k according to Eq. (2);
    x_{m,k} ← x_q ⊙ m_k + x_i ⊙ (1 − m_k);
end
Aggregate all x_{m,k} and m_k to form X_m and M_out;
return X_m, M_out;
```

where N is the number of samples in the current batch. The loss L_i of each sample i is defined as

$$L_i = \frac{-1}{|\mathcal{P}(i)|} \sum_{j \in \mathcal{P}(i)} \log \left(\frac{\exp\left(S'_{ij}/\tau\right)}{\sum_{k=1, k \neq i}^{N} \exp\left(S'_{ik}/\tau\right)} \right) \tag{4}$$

where τ is the temperature parameter and $\mathcal{P}(i)$ denotes the positive sample set corresponding to anchor i. S'_{ij} (and similarly S'_{ik}) denotes the weighted similarity used in the contrastive computation, which is derived from the original similarity S_{ij} by applying modifications exclusively to specific causal pairs (i.e., the original query sample and its corresponding augmented view). The S_{ij} and S'_{ij} are defined as

$$S_{ij} = x_i^{l2} \cdot x_j^{l2} \tag{5}$$

$$S'_{ij} = \begin{cases} w_j \cdot S_{ij} & if \left(\hat{x}_i, \hat{x}_j\right) \text{is the jth causal pair} \\ S_{ij} & otherwise \end{cases} \tag{6}$$

where x_i^{l2} is obtained by $L2$ normalization on all feature vectors $\hat{x}_i$, w_k is the invariance score of the kth query sample, calculated from the corresponding intervention mask matrix m_k

$$w_k = \frac{1}{H \times W} \sum_{u=1}^{H} \sum_{v=1}^{W} (m_k)_{uv} \tag{7}$$

where m_k denotes the mask matrix that is adopted in the RMCI module for the kth query sample, with a size of H $\times$ W.

C. Spatial–Spectral Feature Extraction Network

In our model, we propose a dual-branch spatial–spectral feature extraction network (G) as illustrated in Fig. 3. Spectral branch: the input HSI patch $x \in R^{1 \times B \times H \times w}$ (B is the spectral bands and H and W are the height and width of the patch) is processed by a 3-D convolution with a kernel size of $7 \times 1 \times 1$ and a stride of $3 \times 1 \times 1$ to obtain 24 initial feature maps. Next, a spectral residual block with two 3D-CNN layers ($3 \times 3 \times 3$

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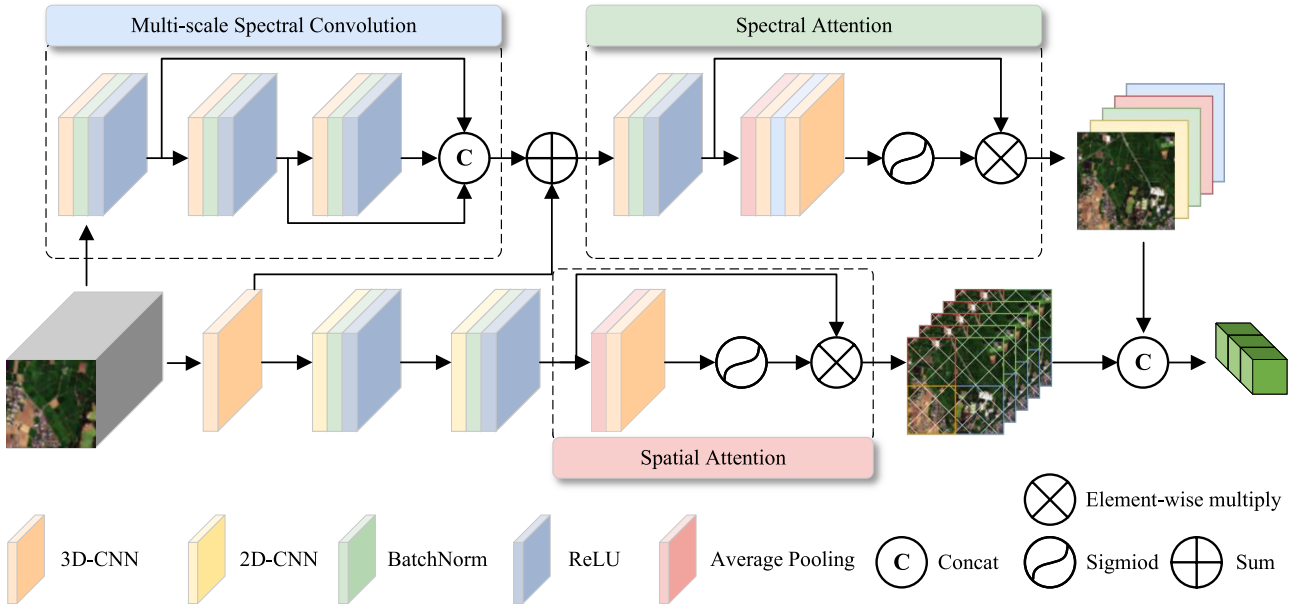


Fig. 3. Architecture diagram of the proposed spatial–spectral network. The framework consists of two parallel branches designed to extract complementary features. The spectral branch utilizes a multiscale spectral convolution module which employs 3D-CNNs, BatchNorm, and ReLU to capture multiscale spectral features. Next, the obtain spectral features are refined through a spectral attention module. The spatial branch processes the input using a series of 2D-CNNs followed by a spatial attention module to enhance discriminative spatial regions. Finally, the outputs from the spectral and spatial attention modules are fused through a concatenation operation to form the final feature representation.

kernels) is followed by a bottleneck 3D-CNN layer that compresses the spectral dimension to 1, which produces 128 mid-level spectral features. We further enhance informative bands with a spectral attention module that consists of the adaptive average pooling operation followed by two 2D-CNN layers with a reduction ratio of 4 to generate a channel-wise weight map, which is applied through element-wise multiplication. Spatial branch: In parallel, a 3D-CNN layer with a $B \times 3 \times 3$ kernel aggregates spatial information into 24 spatial feature maps. A spatial residual block with two 3D-CNN layers (kernel size of $3 \times 3 \times 3$) preserves spatial resolution. Afterward, a global spatial attention module computes an attention map via adaptive global average pooling and a convolution, emphasizing relevant pixels while suppressing background interference. Fusion: The 128-channel spectral features and 24-channel spatial features are concatenated to form a 152-D representation. Finally, a multifeature fusion strategy and an adapter module refine the fused features before the final global average pooling.

D. Collaborative Domain Alignment

In this article, we propose the CDA module, a unified framework designed to bridge domain shifts through a multifaceted training strategy. Unlike independent alignment methods, the CDA integrates causal meta-learning with conditional adversarial training to yield a robust feature space for cross-domain few-shot HSI classification under limited sample conditions.

1) Causal Meta-Learning Branch: In our causal meta-learning branch, initially, we operate on the Q to generate a new M composed of counterfactual samples. Afterward, the three sets, including S, Q, and M, are fed into the feature extractor for meta-training to obtain the features $\hat{x}_s$, $\hat{x}_q$, and $\hat{x}_m$. Furthermore, the class prototypes c_k of the S are computed and the Euclidean metric is employed to measure the similarity between the c_k and the $\hat{x}_m$. Finally, the FSL training loss L_{fsl} is computed for the M in SD and TD, respectively. The c_k of the S is defined as follows:

$$c_k = \frac{1}{|S_k|} \sum_{(\hat{x}_i, y_i)} \hat{x}_i \tag{8}$$

where S_k denotes the set of all samples belonging to class k in the S, $|S_k|$ represents the number of samples in class k, $\hat{x}_i$ is the feature of the ith sample, and y_i is the label of the ith sample.

We define the focal loss as the FSL training loss, as shown in the following equation:

$$L_{fsl} = E_{S,M}\Big[-\sum_{(x_m, y_{true}) \in M_k} \Big(1 - P\Big(y_{i,pred} = y_{true} \mid x_{m,i}\Big)\Big)^{\gamma} \log P\left(y_{i,pred} = y_{true} \mid x_{m,i}\right)\Big] \tag{9}$$

where S denotes the support set, M is the mask set, $y_{i,pred}$ is the prediction label of the sample feature $x_{m,i}$ of the M, and y_{true} is the true label of $x_{m,i}$. The class distribution of the sample $x_{m,i}$ is defined as shown in the following equation:

$$P\left(y_m = k | x_{m,i} \in M\right) = \frac{exp\left(-d\left(G\left(x_{m,i}\right), c_k\right)\right)}{\sum_{k=1}^{c_t} exp\left(-d\left(G\left(x_{m,i}\right), c_k\right)\right)} \tag{10}$$

where $d(,)$ denotes the Euclidean metric, c_k is the class prototypes of the kth class in the S, c_t is the number of S classes, $G(\cdot)$ represents the feature extraction network, $x_{m,i}$ is a the ith sample of the M, and y_m is the true label of $x_{m,i}$.

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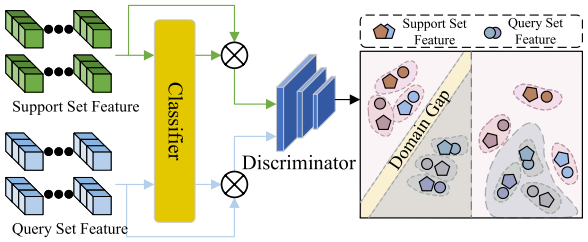


Fig. 4. Schematic of the conditional adversarial training branch. The module addresses the domain gap between the SD and TD, which utilizes a discriminator to differentiate between domains based on conditional features. Adversarial training forces the feature extractor to produce domain-invariant representations, effectively aligning the feature distributions of the support and query sets.

2) Conditional Adversarial Training Branch: In cross-domain few-shot HSI classification, domain-level feature alignment without category constraints often hinders the misalignment of class-level distributions, to achieve effective knowledge transfer and alignment between the SD and TD, we present a conditional adversarial training branch as shown in Fig. 4, which adopts soft probability distributions instead of hard labels to provide the continuous signal, and serves as conditional information for implementing a category-aware domain alignment mechanism. With the S and the Q of the meta-tasks, the conditional adversarial training branch achieves global alignment and transfers effective knowledge in the different HSI domain, which consist of generating conditional features g and a domain discriminator. To form the g, we input the obtained features from the S and Q into the classifier and get the classifier prediction (c) by softmax. The g is defined as follows:

$$g_i = \hat{x}_i \otimes c_i \tag{11}$$

where g_i is the ith conditional features and c_i is the classifier prediction of $\hat{x}_i$. Afterward, the domain discriminator is trained to distinguish whether g belongs to the SD or TD, while the feature extractor aims to produce domain-invariant features that deceive the discriminator. In this way, adversarial training reduces the overall distribution discrepancy between SD and TD. The loss of adversarial training is defined as follows:

$$\begin{aligned}\min_{D}\max_{G,C} L_{\text{adv}} = &-E_{x_i^s\sim P_s(x)}\log\left[D\left(g_i^s\right)\right] \\ &- E_{x_j^t\sim P_t(x)}\log\left[1-D\left(g_j^t\right)\right]\end{aligned} \tag{12}$$

where D is a domain discriminator, G is the feature extractor, C denotes the linear classifier, $P_s(x)$ and $P_t(x)$ represent the data distribution in the SD and TD, $D(\cdot)$ denotes the predicted probability that g_i^s comes from the SD, $1 - D(\cdot)$ denotes the predicted probability that g_i^t comes from the TD, and E indicates expectations.

Overall, the total training loss for CCCDA is defined as follows:

$$L = L_{fsl} + \lambda_1 L_{con} + \lambda_2 L_{adv} \tag{13}$$

where λ_1 and λ_2 are weighting coefficients.

In the early training stages, the classifier predictions c_i may not converge, posing a risk of introducing noise when used as conditional information for adversarial training. The proposed framework mitigates the risk through the synergy of L_{con} and L_{fsl}. The intervention-invariant contrastive loss L_{con} operates directly in the feature space, independent of the classifier outputs. Meanwhile, the classification loss L_{fsl} computes support set prototypes and provides stable supervision using Euclidean distance and focal loss. As L_{con} and L_{fsl} stabilize the feature representations and classification predictions, the reliable conditional probabilities guide the adversarial training. The collaborative mechanism ensures that adversarial training minimizes the domain gap and avoids class confusion between the source and TDs, preserving clear interclass boundaries.

Algorithm 2 Training Phase of CCCDA

Input:
The source dataset D_s;
The target dataset D_t;
The support set S_s and query set Q_s from D_s;
The support set S_t and query set Q_t from D_t;
Number of training episodes N;
hyperparameters λ_1, λ_2, Learning rate=0.001 and Disturbance ratio σ.
Output: The total loss $Loss$ of the model.

```
for n = 1 to 10 do
    for m = 1 to N do
        x_s, x_q ← Divide(x̂);
        x_i ← Sample from x_q;
        Generate a mask matrix m according Equ. (2);
        x_m ← x_q ⊙ m + x_i ⊙ (1 − m) according to Equ. (1);
        x̂_s, x̂_q, x̂_m ← G_θ(x_s, x_q, x_m);
        Calculate prototype features by Equ. (8);
        Calculate L_fsl by Equ. (9);
        Calculate L_con, L_adv by Equ. (3) and (12);
        Calculate overall loss by Equ. (13);
        Loss.backward();
    end
end
```

Algorithm 2 outlines the training procedure for our proposed CCCDA for the task of cross-domain few-shot HSI classification.

III. Experiments

A. Experimental Datasets

We evaluate the performance of our proposed method on four common HSI datasets. In Fig. 5, the detailed parameters of the SD and TD datasets are presented in detail.

1) SD Dataset: Chikusei: We select the Chikusei dataset as the SD. The Chikusei dataset was captured by the Headwall Hyperspec-Vnir-C imaging sensor in agricultural and urban areas in Chikusei, Ibaraki, Japan. This dataset contains 128 spectral bands covering a wavelength range of 363–1018 nm. The scene consists of 2517 × 2335 pixels with a ground sampling distance (GSD) of 2.5 m, and ground truth labels are collected for 19 classes.

2) TD Datasets: a) Pavia University (UP) Dataset: The UP dataset was acquired in 2003 by the reflective optics spectrographic image system (ROSIS) sensor in Pavia, Italy.

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The UP contains nine classes. After removing 12 noisy bands, the dataset comprises 103 usable spectral bands and has a spatial size of 610 × 340 pixels with a spatial resolution of 1.3 m/pixel.

b) Indian Pines (IP) Dataset: The IP dataset was acquired by the AVIRIS sensor over the IP test site in northwestern Indiana. The IP comprises 145 × 145 pixels with a spatial resolution of 20 m/pixel, encompassing 16 land cover classes and 200 spectral bands spanning a wavelength range of 400–2500 nm.

c) Salinas (SA) Dataset: The SA dataset was collected by the AVIRIS sensor over the SA Valley test site in California. The SA consists of 512 × 217 pixels with a spatial resolution of 3.7 m/pixel. The dataset contains 16 land cover classes and 204 spectral bands spanning a wavelength range of 400–2500 nm.

d) Houston2013 (HT) Dataset: The HT dataset was introduced for the 2013 GRSS Data Fusion Contest. The HT dataset was collected in 2012 by an airborne sensor over the HT University campus and surrounding areas. The dataset comprises 349 × 1905 pixels with a spatial resolution of 2.5 m/pixel. HT includes 15 land cover classes and 144 spectral bands spanning a wavelength range of 364–1046 nm.

TABLE I
PARAMETER TUNING OF THE MASK RATIO σ USING THE TD DATASETS

Dataset	Mask ratio σ		
	0.25	0.5	0.75
UP	85.39	**86.83**	86.48
IP	80.90	**81.14**	81.08
SA	91.87	92.36	**92.41**
HT	81.18	**81.31**	80.63

TABLE II
PARAMETER TUNING OF THE L_{con} WEIGHT λ_1 USING THE TD DATASETS

Dataset	L_{con} Weight λ_1			
	0.1	0.5	1	1.5
UP	86.83	86.69	**87.46**	86.97
IP	81.14	**81.76**	81.73	81.10
SA	92.41	92.83	92.62	**92.91**
HT	81.31	81.36	81.31	**81.48**

B. Experimental Setups

1) Hardware Configuration: The workstation adopted in the experiments is equipped with a 10-core Intel[1] Xeon[1] Gold 5418Y CPU, 120-GB RAM, and an NVIDIA RTX 4090 GPU (24-GB VRAM). The experimental code is implemented by PyTorch 1.10.1, CUDA 11.1, and Python 3.8.

2) Experimental Settings: During the meta-training phase, we decompose the entire HSI classification task into many C-way K-shot classification tasks, where C is set as the number of classes of TD (9 for the UP, 15 for the HT, and 16 for the IP and SA), and each class contains K samples. The number of per category for the S is set to 1 and the number of per category for the Q is set to 19. In theory, the high the value of the number of per categories for the Q, the better the model simulates the classification task of the test set, but the limitations of computational efficiency and manually labeled samples must be considered [33]. Notably, there are 19 classes in SD ($C_s > C_t$) and C is randomly extracted from SD when constructing a meta-task. Furthermore, five labeled samples are selected from each class of TD for TD-FSL in all experiments. However, five labeled samples are few for S and Q of TD to be constructed, so the data is augmented by adding Gaussian random noise. In the testing phase, D_{tra} and D_{tes} are treated as S and Q, respectively, and we employ the K-NNs classifier for a classification task. In terms of other parameters, the number of bands and the patch size of the input HSI sample are set to 100 and 9 uniformly, the training iterations were set to 10 000, and the learning rate was set to 0.001 to train the CCCDA sufficiently. Adam is adopted as the optimizer. All convolutional layers and linear layers are initialized by Xavier normalization, and the batch norm layers use the normal distribution $N(1, 0.02)$. Meanwhile, the weight decay is set to $1e^{-4}$ for the stable training of the network.

[1]Registered trademark.

C. Parameter Tuning

To evaluate the impact of different hyperparameter configurations on the performance of the proposed CCCDA framework, we conduct sensitivity analyses on three key parameters, including the mask ratio in the RMCI module, the intervention-invariant contrastive loss weight, and the adversarial training loss weight. All experiments are performed across four TD datasets under the C-way 5-shot setting.

1) *Impact of Mask Ratio σ:* The mask ratio determines the intensity of the causal intervention by controlling the proportion of noncausal background pixels being replaced. As shown in Table I, the classification accuracy generally improves as increases from 0.25 to 0.50, while the performance tends to decline when reaches 0.75. Specifically, for the UP, IP, and HT datasets, the optimal performance is achieved at = 0.50, where the model yields accuracies of 86.83%, 81.14%, and 81.31%, respectively. The results verify that a moderate disturbance ratio (50%) effectively suppresses spurious correlations without overwhelming the model with excessive noise.
2) *Impact of L_{con} Weight λ_1:* The parameter λ_1 balances the contribution of the intervention-invariant contrastive learning loss, which enforces feature invariance against causal interventions. Table II illustrates the tuning results for λ_1. For the UP dataset, the accuracy peaks at 87.46% when λ_1 = 1.0. For the IP dataset, a smaller weight (λ_1 = 0.5) yields the best result of (81.76%), while for the SA and HT datasets, the optimal performance is shifted toward a larger weight when λ_1 equals to 1.5. The results indicate that datasets with higher intraclass variance may require a stronger contrastive signal to learn stable and causally-related features.

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TABLE III
PARAMETER TUNING OF THE L_{adv} WEIGHT λ_2 USING THE TD DATASETS

Dataset	L_{adv} Weight λ_2			
	0.01	0.1	0.5	1
UP	86.89	**87.46**	87.24	87.29
IP	81.00	**81.76**	80.75	79.14
SA	92.91	92.33	93.01	**93.39**
HT	81.41	**81.48**	80.85	80.24

3) *Impact of L_{adv} Weight λ_2:* The weight λ_2 regulates the conditional adversarial training branch, which is responsible for bridging the domain gap through global distribution alignment. As reported in Table III, the UP, IP, and HT datasets achieve their highest accuracies (87.46%, 81.76%, and 81.48%) at $\lambda_2 = 0.1$. In contrast, the SA dataset requires a higher weight ($\lambda_2 = 1.0$) to reach its peak performance of 93.39%. The results highlight that the magnitude of the domain shift varies across different target scenes, requiring adaptive adjustment of the adversarial loss to ensure effective knowledge transfer from the SD.

D. Comparison With State-of-the-Art (SOTA) Methods

To assess the classification performance of the proposed CCCDA framework in limited labeled samples, we compare with six SOTA methods: Gia-CFSL [45], ADAFSL [46], FDFSL [52], CTF-SSCL [53], GCC-FSL [54], and MLPA [55]. For a fair comparison, identical hardware parameters, random seed, batch size, and train and test classes are applied in all methods. The Chikusei dataset is adopted as the SD dataset for HSI classification methods, and apply the same sampling strategy. And to mitigate the impact of random sampling, all methods are performed ten independent repetitions and the experimental results of are averaged. The remaining settings employ the default values from the open-source code and the original paper. In addition, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are selected as evaluation metrics in this article. The highest classification accuracy is shown in bold and the second-highest accuracy is underlined.

Tables IV–VII present the classification results of all methods in the UP, IP, SA, and HT datasets with five labeled samples randomly selected from each target class. Overall, in the three evaluation metrics OA, AA, and Kappa, the CCCDA method outperformed Gia-CFSL, ADAFSL, FDFSL, CTF-SSCL, GCC-FSL, and MLPA on all four datasets.

1) Classification Maps and Results of the UP Dataset: The primary challenge of the UP dataset is the high spectral similarity between urban classes. The MLPA method aims to reduce domain shift by aligning class prototypes at various network levels. While multilevel prototype alignment is an effective strategy, the MLPA approach operates on features that may already be corrupted by spurious correlations. The limitation is evident in the low 61.84% accuracy of MLPA for class 3. In contrast, our CCCDA employs a causal intervention module to mitigate spurious correlations before domain alignment, achieving 83.10% accuracy on the same class. The classification map in Fig. 6 visually confirms the lower error rate and superior structural consistency of the model.

2) Classification Maps and Results of the SA Dataset: For the SA dataset, characterized by complex textures and high intraclass variance, the Gia-CFSL method addresses spatial complexity by leveraging graph networks to aggregate nonlocal information. The graph aggregation process combines features from a central node with spatial proximity nodes. But, these spatial proximity nodes often consist of a mix of same-class pixels and varied background pixels. The feature blending degrades the performance of Gia-CFSL, resulting in an OA of only 88.48%. Our CCCDA, by separating causal features from the noisy contextual information, achieves a high precision in 13 of 16 classes (see Table VI). Fig. 7 illustrates the ability of the model to maintain clear class boundaries in visually complex areas.

3) Classification Maps and Results of the IP Dataset: The IP dataset presents challenges of severe class imbalance and spectral confusion among vegetation types. The feature disentanglement in FDFSL is proposed to separate domain-specific from domain-invariant statistical properties. However, FDFSL focuses on statistical independence rather than explicit causal relationships, and thus may still group causally-spurious features within the domain-invariant representation. Consequently, the proposed CCCDA surpasses FDFSL in OA by a significant 12.07%, as detailed in Table V. The proposed causal framework constructs a more robust feature space, leading to fewer misclassified pixels in spectrally confusing regions (see Fig. 8).

4) Classification Maps and Results of the HT Dataset: The HT dataset introduces the challenge of large intraclass variability and intricate urban structures. For this challenge, the CTF-SSCL fuses CNN and transformer features to enhance classification precision. The CTF-SSCL relies on a spectral feature shift strategy for augmentation to improve class consistency. The feature augmentation strategy may reinforce noncausal, spurious features, limiting the overall adaptability. A performance decline is reflected in the lower accuracies for CTF-SSCL on high-variance classes 7 and 9. In contrast, our CCCDA proposes CICL and CDA to mitigate such problems, achieving high accuracies on the same classes (see Table VII). The classification results in Fig. 9 demonstrate spatial continuity and class-wise consistency.

To mathematically substantiate the superiority of the proposed framework, a rigorous pairwise statistical significance test is integrated into the evaluation protocol. Specifically, the Wilcoxon signed-rank test is adopted to compare the OA results between the proposed CCCDA and the most competitive baseline across the ten independent runs. The CCCDA framework was evaluated against the FDFSL method on the UP dataset, the IP dataset, and the SA dataset, and against the CTF-SSCL method on the HT dataset. The calculated exact p values yield 0.01367 for the UP dataset, 0.00098 for the IP dataset, and 0.00684 for the SA dataset. These consistently low p values confirm that the performance gains achieved by the CCCDA framework on these datasets are statistically significant. For the HT dataset, the p value is

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TABLE IV

CLASSIFICATION ACCURACY (%) OF VARIOUS METHODS ON THE UP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	Gia-CFSL	ADAFSL	FDFSL	CTF-SSCL	GCC-FSL	MLPA	Our
1	81.74±11.63	85.12±3.29	80.13±7.12	87.99±4.34	78.67±10.32	76.60±16.29	**95.41±3.13**
2	84.83±6.54	77.62±12.51	84.56±7.40	78.53±8.22	85.46±7.74	**88.39±5.13**	82.92±11.95
3	61.26±12.33	67.43±12.71	64.97±8.89	66.96±8.86	59.26±11.88	61.84±10.31	**83.10±10.98**
4	90.69±5.21	88.51±2.27	91.06±4.55	88.01±3.82	**93.89±3.41**	92.52±3.22	92.93±4.24
5	98.13±2.46	98.91±1.26	99.44±0.54	**99.86±0.31**	98.81±1.14	99.29±1.36	99.51±0.54
6	70.45±9.21	77.31±15.27	70.93±11.37	75.17±10.99	74.23±6.71	73.65±8.28	**84.87±10.15**
7	75.72±10.06	88.85±2.46	85.75±7.19	90.51±7.31	79.71±10.38	77.69±8.63	**96.54±3.67**
8	72.27±11.96	79.28±10.00	81.73±8.67	81.60±12.65	70.47±14.45	62.15±13.40	**87.25±4.02**
9	97.45±4.01	98.87±1.35	92.92±6.47	93.06±8.75	**98.97±1.07**	98.81±1.42	97.94±3.40
OA	81.26±2.25	80.65±4.55	82.22±2.95	80.99±4.01	81.66±3.10	81.81±2.26	**87.46±4.75**
AA	81.39±2.02	84.66±1.87	83.50±1.68	84.63±2.56	82.16±1.95	81.22±1.49	**91.16±2.25**
Kappa	75.66±2.61	75.26±5.07	76.91±3.50	76.12±4.44	76.27±3.62	76.30±2.75	**83.92±5.50**

Note: The p-value is 0.01367 (Our vs. FDFSL) based on the Wilcoxon signed-rank test.

TABLE V

CLASSIFICATION ACCURACY (%) OF VARIOUS METHODS ON THE IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	Gia-CFSL	ADAFSL	FDFSL	CTF-SSCL	GCC-FSL	MLPA	Our
1	94.39±5.57	96.83±4.50	97.56±2.89	97.07±4.60	93.66±8.74	94.39±12.10	**99.76±0.73**
2	44.67±11.98	54.68±12.29	53.13±7.26	53.35±9.83	44.95±11.25	48.83±11.00	**74.60±11.37**
3	48.79±12.05	63.27±6.36	63.31±8.69	57.92±10.72	54.96±5.21	46.13±11.22	**71.68±8.58**
4	78.71±7.88	73.97±15.24	84.44±11.32	84.01±10.69	77.11±11.69	74.57±14.97	**96.16±7.11**
5	74.52±8.28	79.90±8.79	71.53±11.46	78.95±8.29	72.68±7.09	75.02±8.97	**82.59±6.91**
6	82.79±8.33	84.18±7.06	87.41±7.66	90.66±5.65	84.17±8.57	87.27±5.57	**95.92±2.12**
7	100.00±0.00	98.70±2.78	99.13±1.74	99.57±1.30	97.39±6.51	97.39±6.51	**100.00±0.00**
8	88.92±9.25	87.55±13.32	85.37±10.40	92.94±8.31	89.51±9.87	86.96±7.37	**95.18±7.08**
9	100.00±0.00	100.00±0.00	100.00±0.00	97.33±6.11	100.00±0.00	100.00±0.00	**100.00±0.00**
10	55.90±11.23	66.24±7.80	63.90±11.30	66.42±10.51	57.63±11.18	63.70±6.18	**75.44±9.73**
11	61.36±12.10	59.96±16.35	62.87±5.79	59.33±15.85	60.10±11.08	60.49±12.62	**75.90±8.09**
12	44.73±11.08	57.84±16.89	54.69±9.71	54.32±13.16	44.97±13.03	49.80±15.00	**78.23±17.29**
13	**98.95±1.01**	97.75±2.88	97.65±3.38	98.80±1.47	98.60±2.03	96.30±5.65	98.95±1.75
14	79.13±9.58	83.29±11.70	86.68±6.66	84.56±10.98	80.86±10.09	88.37±6.29	**91.90±7.70**
15	63.12±11.17	73.04±11.39	79.11±14.54	80.00±11.25	67.53±10.34	72.97±13.23	**85.67±12.24**
16	97.73±2.64	99.09±0.85	98.18±2.88	98.64±3.04	98.98±2.06	98.30±3.02	**99.20±1.69**
OA	63.95±2.81	68.99±3.72	69.69±1.63	69.36±3.51	64.74±2.40	66.76±2.61	**81.76±2.40**
AA	75.86±1.86	79.77±1.73	80.31±1.78	80.87±1.34	76.44±1.63	77.53±1.57	**88.82±1.48**
Kappa	59.29±3.07	65.13±3.91	65.95±1.76	65.54±3.53	60.23±2.55	62.58±2.70	**79.36±2.74**

Note: The p-value is 0.00098 (Our vs. FDFSL) based on the Wilcoxon signed-rank test.

TABLE VI

CLASSIFICATION ACCURACY (%) OF VARIOUS METHODS ON THE SA DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	Gia-CFSL	ADAFSL	FDFSL	CTF-SSCL	GCC-FSL	MLPA	Our
1	98.53±2.37	99.34±0.84	98.70±1.41	99.32±1.21	99.45±0.71	98.42±1.73	**99.87±0.20**
2	99.32±0.93	99.68±0.32	99.14±1.88	99.37±1.12	99.37±0.85	99.55±0.43	**99.99±0.04**
3	92.77±6.66	**93.73±7.97**	90.54±11.63	91.34±11.81	91.81±4.85	89.54±11.44	92.81±9.69
4	99.37±0.74	98.70±1.50	99.25±0.61	99.01±1.20	99.47±0.76	98.96±0.97	**99.88±0.20**
5	88.82±7.00	**95.37±1.84**	90.94±7.73	92.94±6.05	89.54±3.57	92.29±3.72	95.19±3.53
6	98.65±1.70	98.97±0.97	98.57±1.36	99.18±0.83	98.59±1.59	98.92±1.26	**99.37±0.91**
7	97.52±2.92	99.33±1.15	99.05±1.19	99.35±0.85	96.51±7.99	98.80±1.73	**99.45±0.75**
8	76.27±12.48	73.93±10.10	80.06±4.33	73.51±11.93	74.93±14.42	77.32±10.40	**86.03±8.42**
9	97.57±3.02	99.73±0.31	98.39±2.08	98.78±1.63	97.43±3.42	99.52±0.85	**99.77±0.58**
10	82.01±6.73	84.39±6.53	81.47±9.76	84.74±7.69	83.39±8.45	85.53±8.34	**91.45±3.55**
11	97.06±2.32	98.08±1.49	97.94±2.29	96.82±1.62	96.68±2.40	97.80±1.67	**99.82±0.19**
12	98.66±1.90	99.73±0.28	98.89±1.15	98.70±1.09	98.80±2.18	99.28±0.46	**99.03±1.51**
13	98.08±1.89	98.72±2.18	97.42±2.08	98.46±1.84	97.76±2.06	99.12±0.73	**99.62±0.53**
14	97.50±1.50	97.80±2.57	97.93±2.58	97.62±1.60	98.52±0.99	98.99±0.78	**98.92±2.19**
15	74.35±8.45	79.32±4.92	**84.06±6.46**	77.14±9.18	74.92±6.91	76.64±6.66	83.03±6.46
16	88.95±6.15	90.47±5.01	90.53±8.10	91.20±6.42	91.00±5.01	92.33±4.99	**93.51±3.89**
OA	88.48±2.17	89.72±2.19	90.83±1.49	89.01±2.54	88.40±2.85	89.81±1.90	**93.39±1.62**
AA	92.84±1.02	94.20±0.95	93.93±1.26	93.59±1.30	93.01±1.21	93.94±1.15	**96.11±0.85**
Kappa	87.21±2.39	88.60±2.41	89.81±1.65	87.81±2.64	87.13±3.12	88.68±2.09	**92.65±1.79**

Note: The p-value is 0.00684 (Our vs. FDFSL) based on the Wilcoxon signed-rank test.

0.09668, indicating a strong positive trend toward significance at the 0.10 level. The corresponding significance levels have been incorporated into the overall performance tables using standard asterisk markers.

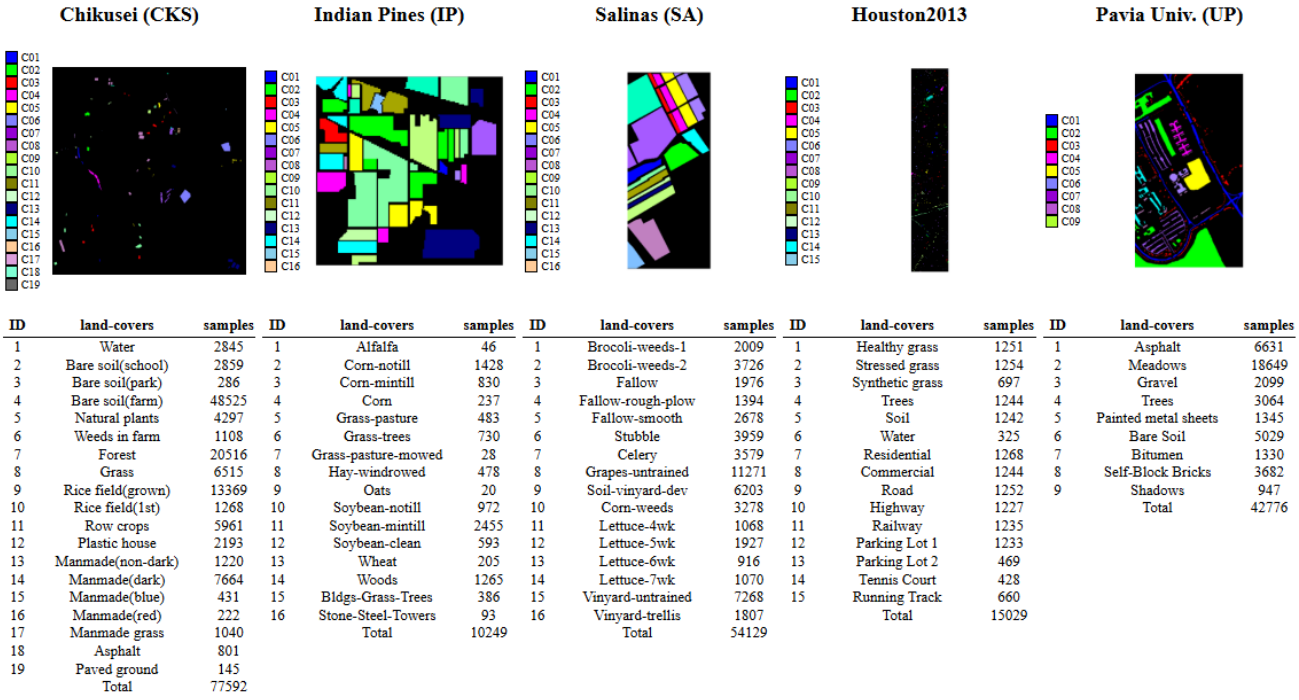
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TABLE VII

CLASSIFICATION ACCURACY (%) OF VARIOUS METHODS ON THE HT DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	Gia-CFSL	ADAFSL	FDFSL	CTF-SSCL	GCC-FSL	MLPA	Our
1	88.77±6.53	89.43±6.09	86.85±5.36	89.49±6.37	**89.82±5.92**	87.99±5.69	89.56±6.07
2	78.49±9.89	85.99±9.61	89.02±5.26	87.29±10.20	82.89±9.82	82.90±6.48	**89.87±7.01**
3	95.97±5.76	97.86±3.22	97.43±2.23	98.38±1.82	95.85±3.17	96.76±2.63	**99.22±1.03**
4	90.40±4.09	90.22±3.66	91.40±3.37	92.78±3.29	92.30±3.29	91.13±2.00	**94.12±2.66**
5	94.31±4.69	97.59±2.81	**99.22±0.85**	94.36±6.23	96.39±2.79	96.60±4.19	99.20±0.92
6	73.34±8.93	77.69±8.07	82.94±7.09	**87.16±6.59**	75.12±8.14	79.03±9.15	86.84±7.25
7	67.55±8.42	59.40±11.47	69.88±9.44	72.31±9.51	58.24±14.90	52.64±13.39	**82.95±8.12**
8	**44.72±12.05**	40.48±4.23	48.91±9.07	45.05±8.68	42.59±8.56	46.88±9.11	42.15±7.69
9	58.12±12.95	64.61±9.34	69.87±8.09	75.30±7.07	60.44±14.84	58.94±10.61	**78.71±8.25**
10	68.11±11.24	66.99±17.74	61.37±15.52	**69.63±14.95**	66.85±10.91	68.90±15.53	68.38±20.21
11	63.20±10.03	64.91±7.55	68.07±7.94	67.91±11.18	63.70±13.88	55.64±13.25	**72.91±11.84**
12	70.81±15.34	69.52±12.32	62.93±11.10	**73.59±11.90**	70.92±11.03	72.23±11.71	65.78±15.35
13	89.85±5.57	45.24±16.58	**95.39±1.86**	90.00±4.91	87.96±6.36	87.78±9.61	90.60±4.89
14	88.75±7.22	94.92±4.54	96.60±4.60	94.82±6.09	94.00±3.41	91.21±5.93	**99.91±0.28**
15	93.50±4.98	93.60±5.25	98.03±2.97	97.80±3.80	92.29±6.77	96.26±3.37	**99.89±0.32**
OA	75.48±1.84	74.84±2.03	78.29±2.35	79.85±2.22	75.51±1.67	75.71±1.43	**81.48±2.35**
AA	77.73±1.23	75.90±2.45	81.19±1.78	82.39±1.90	77.96±1.15	78.33±1.47	**84.01±1.97**
Kappa	73.52±1.98	72.79±2.21	76.56±2.52	78.24±2.27	73.56±1.78	73.77±1.53	**80.01±2.54**

Note: The p-value is 0.09668 (Our vs. CTF-SSCL) based on the Wilcoxon signed-rank test.



ID	land-covers	samples
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farm)	48525
5	Natural plants	4297
6	Weeds in farm	1108
7	Forest	20516
8	Grass	6515
9	Rice field(grown)	13369
10	Rice field(1st)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	Total	77592

ID	land-covers	samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Bldgs-Grass-Trees	386
16	Stone-Steel-Towers	93
	Total	10249

ID	land-covers	samples
1	Brocoli-weeds-1	2009
2	Brocoli-weeds-2	3726
3	Fallow	1976
4	Fallow-rough-plow	1394
5	Fallow-smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes-untrained	11271
9	Soil-vinyard-dev	6203
10	Corn-weeds	3278
11	Lettuce-4wk	1068
12	Lettuce-5wk	1927
13	Lettuce-6wk	916
14	Lettuce-7wk	1070
15	Vinyard-untrained	7268
16	Vinyard-trellis	1807
	Total	54129

ID	land-covers	samples
1	Healthy grass	1251
2	Stressed grass	1254
3	Synthetic grass	697
4	Trees	1244
5	Soil	1242
6	Water	325
7	Residential	1268
8	Commercial	1244
9	Road	1252
10	Highway	1227
11	Railway	1235
12	Parking Lot 1	1233
13	Parking Lot 2	469
14	Tennis Court	428
15	Running Track	660
	Total	15029

ID	land-covers	samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-Block Bricks	3682
9	Shadows	947
	Total	42776

Fig. 5. Ground truth and the detailed numbers of the samples for each class in the source and target datasets.

E. Ablation Studies

To demonstrate the contribution of the presented models of the CCCDA, we conducted ablation studies on the four datasets, and the average OA, AA, Kappa, and standard deviations of the ten times are shown in Table VIII. In the experiments, the baseline represents the presented network without the CICL and CDA modules. As can be observed, the approach with the baseline generates the lowest OA, AA, and Kappa for the four datasets.

1) Baseline: The basic meta-learning network without the CICL and CDA modules.

2) Baseline + CICL: The network with the CICL module enabled, evaluating the impact of the causal intervention strategy on the model.

3) Baseline + CICL + CDA (Complete Model): The complete CCCDA framework with both the CICL module and CDA modules activated, assessing the joint contribution of the CICL module and CDA module.

As shown in Table VIII, the proposed CICL module continues to improve performance in all datasets, validating the effectiveness of the CICL in facilitating causal-related feature extraction. For example, on the IP dataset, CICL increases OA from 74.21% to 78.81%, AA from 84.39% to 87.47%,

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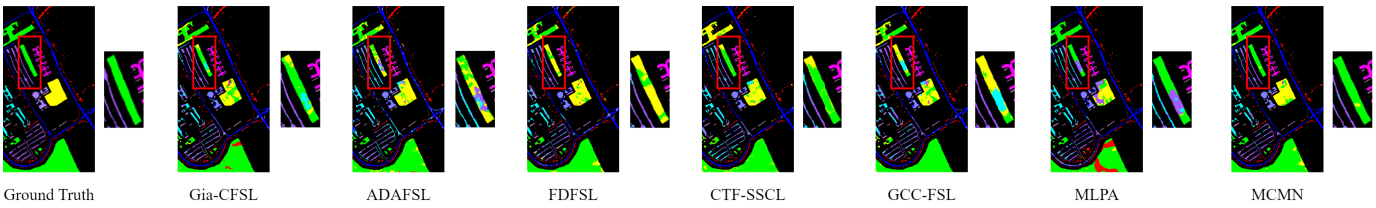


Fig. 6. Classification visualization maps of different models on the UP dataset.

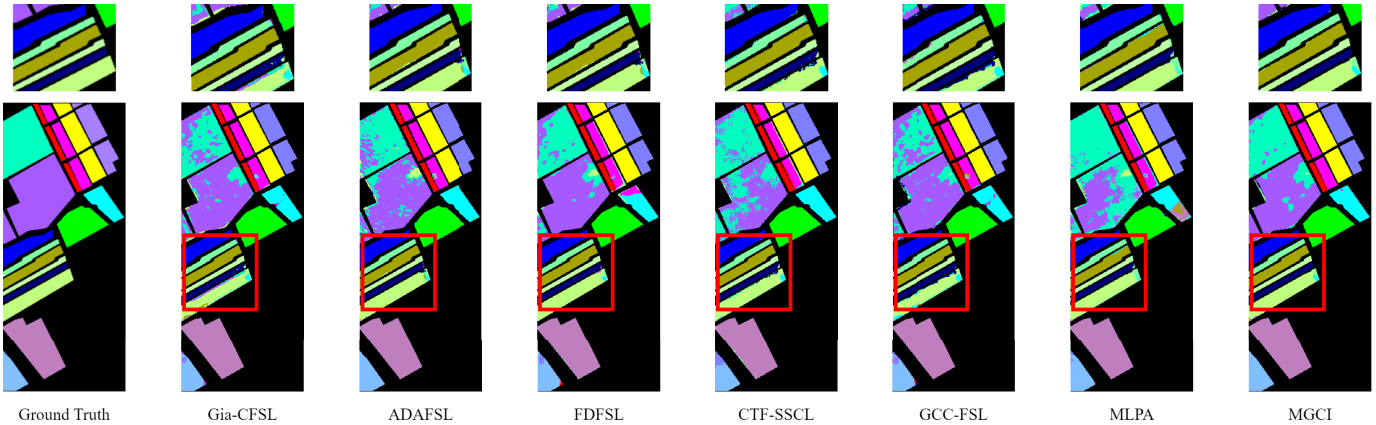


Fig. 7. Classification visualization maps of different models on the SA dataset.

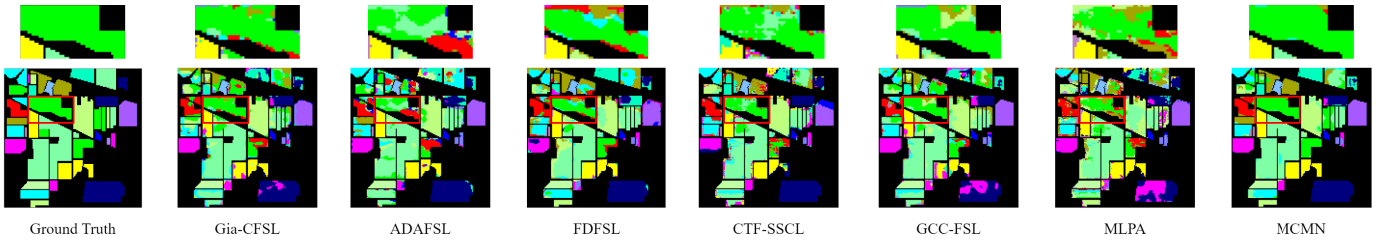


Fig. 8. Classification visualization maps of different models on the IP dataset.

TABLE VIII
ABLATION EXPERIMENT

	Module			Accuracy		
	Baseline	CICL	CDA	OA	AA	Kappa
Pavia University	√			83.31±4.67	87.92±2.38	78.79±5.22
	√	√		86.61±5.28	90.96±2.02	82.95±6.00
	√	√	√	87.46±4.75	91.16±2.25	83.92±5.50
Indian Pines	√			74.21±3.16	84.39±1.33	70.96±3.47
	√	√		78.81±3.80	87.47±1.82	76.12±4.15
	√	√	√	81.76±2.40	88.82±1.48	79.36±2.74
Salinas	√			91.34±1.86	94.96±0.99	90.38±2.04
	√	√		91.82±1.94	95.69±0.97	90.92±2.13
	√	√	√	93.39±1.79	96.11±0.85	92.62±1.79
Houston2013	√			80.93±2.34	83.36±2.22	79.42±2.53
	√	√		81.47±2.31	83.97±1.97	80.00±2.49
	√	√	√	81.48±2.35	84.01±1.97	80.01±2.54

and Kappa from 70.96% to 76.12%. Further integration of the CDA module brings additional performance gains, particularly in scenarios with complex HSI domain gaps. In the UP dataset, the addition of CDA to the CICL-enhanced model leads to an improvement in OA from 86.61% to 87.46% and Kappa from 82.95% to 83.92%, which demonstrates the role of CDA in aligning feature distributions. The above results suggest that CICL and CDA provide complementary benefits, where CICL improves intradomain feature quality through causal intervention, while CDA reinforces cross-domain generalization by minimizing domain shifts. The complete CCCDA model, with both components, achieves the best or near-best performance on all datasets, validating the effectiveness of the proposed framework.

F. Computational Complexity Analysis

In this section, we analyze the computational complexity of the proposed method in terms of training time, testing time, floating-point operations (FLOPs), and the number of model parameters (#Params). Table IX reports a detailed comparison between our method and six SOTA methods on all four HSI datasets. For the #Params and FLOPs, the proposed CCCDA maintains a #Params of approximately 0.41–0.42 M across all datasets, which is slightly higher than baseline methods like ADAFSL and MLPA. The higher cost mainly stems from the dual-branch spatial–spectral extractor and adversarial training components. Similarly, the FLOPs for CCCDA range from 4535.44 to 4617.36 M, which is comparable to Gia-CFSL and GCC-FSL. For the training efficiency, the training time of CCCDA is relatively moderate. CCCDA requires 2849.52 s for training on the UP dataset, which outperforms ADAFSL and is comparable to Gia-CFSL. For the testing efficiency, CCCDA

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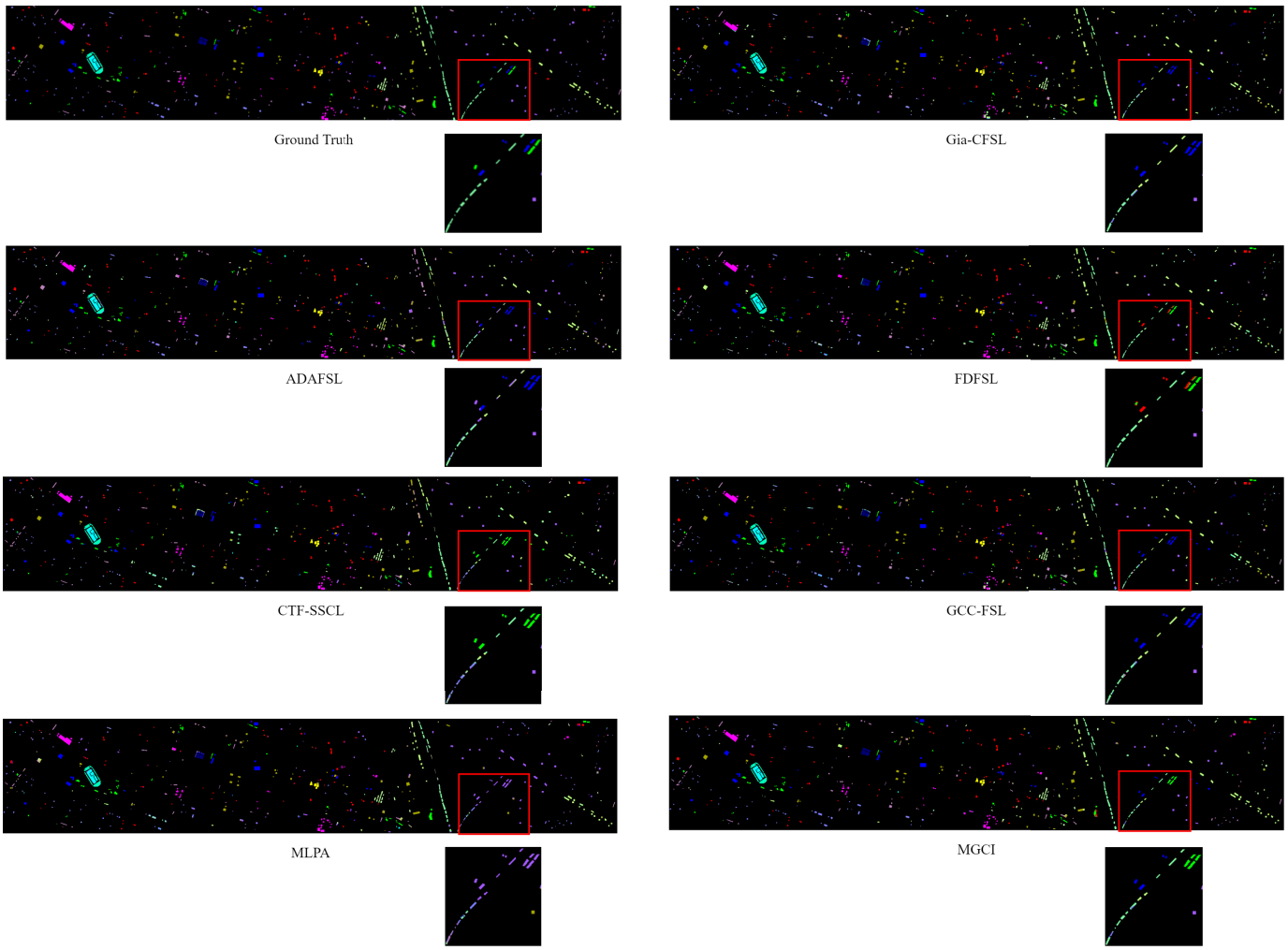


Fig. 9. Classification visualization maps of different models on the HT dataset.

TABLE IX
COMPUTATION TIME (S), FLOPS (M), AND PARAMETERS (M) ON EACH DATASET WITH ALL METHODS

Dataset	Metric	Method						
		Gia-CFSL	ADAFSL	FDFSL	CTF-SSCL	GCC-FSL	MPLA	Our
UP	Training time	3601.51	28254.33	769.24	869.17	2117.83	758.37	2849.52
	Testing time	2.89	3.54	6.35	2.74	2.31	3.58	2.02
	FLOPs	4252.71	2999.01	5193.71	556.37	4252.71	4245.41	4535.44
	#Params	0.20	0.04	0.18	0.08	0.20	0.05	0.41
IP	Training time	4373.14	2277.47	1167.07	2047.83	3739.18	1132.15	4251.43
	Testing time	0.79	0.95	1.62	0.80	0.67	0.64	0.54
	FLOPs	4331.28	3077.58	5272.28	634.96	4331.28	4324.09	4614.12
	#Params	0.21	0.05	0.19	0.09	0.21	0.06	0.42
SA	Training time	4445.01	2393.05	1225.49	2126.13	3825.67	1187.58	5047.99
	Testing time	4.13	5.52	9.96	4.45	3.57	3.13	3.13
	FLOPs	4334.52	3080.82	5275.52	638.20	4334.52	4327.33	4617.36
	#Params	0.21	0.05	0.19	0.09	0.21	0.06	0.42
HT	Training time	3788.51	3218.93	566.40	1445.20	3539.52	1012.56	4105.54
	Testing time	1.14	1.18	1.28	1.07	0.81	0.76	0.66
	FLOPs	4285.92	3032.22	5226.92	589.59	4285.92	4278.71	4568.74
	#Params	0.20	0.04	0.18	0.08	0.20	0.05	0.42

demonstrates superior performance in inference speed. Our method achieves the lowest testing time on the UP, IP, and HT datasets among all compared methods, which indicates that the final feature extractor and classifier are well optimized

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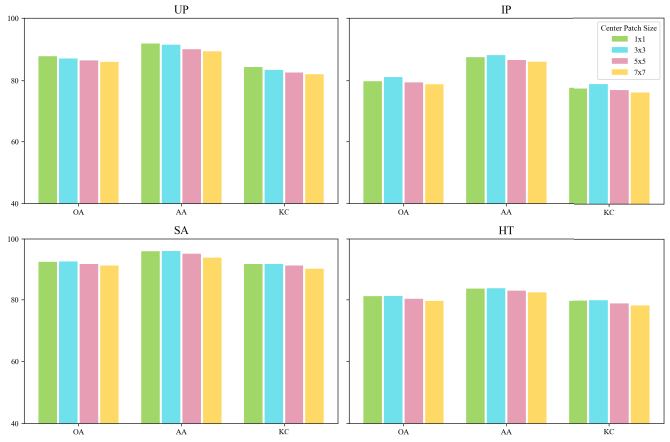


Fig. 10. Performance comparison for the four datasets with different center patch sizes.

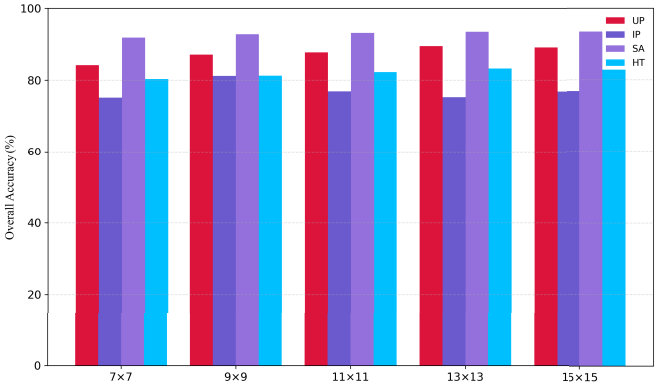


Fig. 11. Performance comparison for the four datasets with different sample patch sizes.

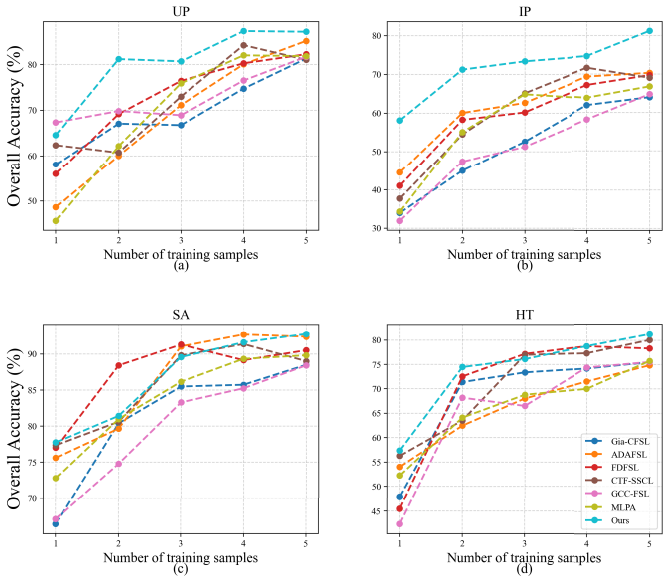


Fig. 12. Classification results with different numbers of labeled samples on target datasets. (a) UP. (b) IP. (c) SA. (d) HT.

for rapid land-cover recognition in TDs. Overall, CCCDA introduces a slight increase in parameters to reduce spurious correlations and domain shifts, while maintaining an excellent balance between computational cost and classification performance.

G. Study of Different Center Patch Size

To evaluate the impact of the central area on the CICL module, the size of the protected central region within each patch is varied from 1×1 to 7×7. As shown in Fig. 10, a clear trend emerges across all four datasets, demonstrating that the classification performance, measured by OA, AA, and Kappa, generally declines as the central preserved region excessively increases. The results from the 1×1 and 3×3 configurations suggest that preserving a minimal, class-relevant core is sufficient for capturing essential causal signals. Specifically, the optimal size of the preserved core is intrinsically tied to the spatial continuity and texture complexity of the specific datasets. The 1×1 setting achieved optimal performance on the UP dataset. The phenomenon is closely related to the spatial characteristics of UP, which is a complex urban scene characterized by highly fragmented land covers, intricate boundaries, and small-scale discrete objects. In such a spatially complex environment, expanding the preserved core to 3×3 risks introducing neighboring pixels of different categories into the causal feature set, thereby corrupting the label purity of the causal intervention.

Conversely, on the IP, SA, and HT datasets, the 3×3 setting demonstrates a slight advantage over the 1×1 configuration. These datasets predominantly feature larger, spatially contiguous regions, such as expansive agricultural tracts or broadly distributed homogeneous land covers. In the domains, a 3×3 neighborhood exhibits high spatial continuity and label consistency. Therefore, preserving a 3×3 core not only maintains label purity but also retains valuable local textural patterns and spatial correlations. Since the label is strictly determined by the center pixel, pixels closer to the center heavily involve the core category characteristics, whereas pixels away from the center are highly vulnerable to interference from neighboring objects or complex mixed pixels. Hence, the RMCI module guarantees purer causal information when the preserved region is constrained to a smaller, adaptive scale.

H. Analyzing of Numbers of Sample and Patch Size

In this article, we analyzed the impact of training sample counts from 1 to 5 and patch sizes from 7×7 to 15×15. As shown in Fig. 11, model accuracy generally improved as patch size increased from 7×7 to 13×13, as a larger sensing field enhances spatial context modeling. However, accuracy decreased at 15×15, particularly on the IP and UP datasets, likely due to interference from irrelevant background regions. While the SA and HT datasets were more stable to patch size changes, the highest accuracy was achieved with a 13×13 patch for the UP, SA, and HT datasets, whereas a 9×9 patch was optimal for the IP dataset. Fig. 12 presents a comparison of the classification results for the proposed CCCDA and six other methods on the target datasets with a varying number of labeled samples (1–5). CCCDA consistently outperformed all

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competitors on all datasets, showing a significant advantage in extremely low-shot settings (1–3 samples). On the complex IP dataset, the accuracy of the CCCDA was nearly 10% higher than the second-best method. The proposed CCCDA demonstrated robust and superior performance on the UP, SA, and HT datasets in all few-shot scenarios.

The results of the above study validate the effectiveness of the proposed approach in cross scene and FSL. The CCCDA consistent lead in diverse HSI datasets suggests that the method is able to capture class-discriminative features and mitigate the adverse impact of distribution shift and label sparsity. The capability is crucial in the cross-domain few-shot HSI classification.

I. Failure Analysis and Limitations

While the proposed CCCDA framework demonstrates superior performance across various HSI datasets, certain limitations and failure cases were observed during extensive testing.

1) In the UP dataset, despite employing intervention-invariant feature extraction, some confusion remains between meadows and bare soil classes. The mentioned issue arises from the highly overlapping spectral signatures in specific bands, which hinders the feature extractor to fully distinguish the classes under limited labeling conditions.
2) In the IP dataset, the proposed model occasionally exhibits sensitivity in fragmented agricultural areas. At the boundaries of different vegetation types, the 3×3 central region of the sample patch may contain mixed pixels, which potentially leads to slight feature representation bias during the counterfactual sample generation process.
3) For the SA dataset, although the model is able to clearly identify various classes, the performance suffers a slight decline when dealing with highly variable intraclass textures. The result indicates that the current RMCI module requires an adaptive mask ratio to effectively confront data with complex textures.
4) For the HT dataset, since it contains complex land cover and has obvious discrepancies from the SD, the proposed model is more sensitive to the coefficient of the adversarial domain-alignment loss than the other three datasets. In this case, the value of λ_2 should be set regarding the target dataset to improve the generalization ability.

IV. Conclusion

In this article, we present a novel causal-intervened contrastive learning with CDA framework for cross-domain few-shot HSI classification, aiming to alleviate performance degradation caused by spectral variability-induced spurious correlations and the scarcity of labeled samples. We present a CICL strategy, where the RMCI module generates a mask set of counterfactual samples to disturb noncausal factors while preserving semantic information. Building upon this, the intervention-invariant contrastive learning branch acts to enforce consistency between the query and the counterfactual views, ensuring the extracted features remain robust to background variations. Furthermore, the CDA module is proposed to bridge the domain shift, utilizing a causal meta-learning branch for prototype alignment and a conditional adversarial training branch for global distribution alignment. The superiority of our method has been demonstrated through extensive experiments on four public HSI datasets. In the future, we plan to investigate the discovery of adaptive causal information and extend this causal framework to multi-SD or multimodal scenarios, which is beneficial for complex scenarios in cross-domain few-shot HSI classification.

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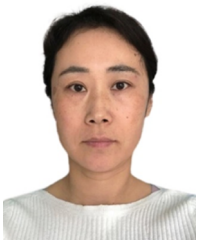
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